alodi

PROJECT REPORT

Make the second drop land.

Alodi is an AI agent that rescues failed deliveries for Vietnamese e-commerce. This report covers the problem, the product, the architecture, the OpenAI stack, the evaluation plan, the economics and the road to a pilot. Every figure carries its source and its status.

₫473M	1 tap	0	₫194M
LOST A MONTH IN ONE AREA TODAY · MODELLED	FROM THE COURIER STARTS A RESCUE	HIGH-STAKES ACTIONS WITHOUT A PERSON	NET SAVING A MONTH, 41% · PROJECTED

CONTENTS

In this report

1. Executive summary
2. The problem
3. What we heard: the survey
4. Alodi
5. The product
6. Architecture
7. OpenAI at the core
8. Data, privacy and rules
9. Evaluation and outcomes
10. Economics
11. Competitive landscape
12. Roadmap
13. Risks and assumptions
14. Team
15. Sources and verification

> Status labels used throughout: **Sourced** public document · **Vendor** company claim · **Estimate** team assumption · **Modelled** sourced numbers combined with stated assumptions · **Projected** not yet measured · **Simulated** fictional mock data.

01

Executive summary

A failed delivery is a time negotiation that happens off the system. Alodi brings it back in.

The problem

Sellers pay up to ₫40,000 in return shipping for every failed Shopee order, 8–12% of orders are refused at the door, and buyers still cannot choose a delivery window. When a courier can't reach a buyer, the context that would fix the next attempt is compressed into one error code. For one area of 20 couriers and 4,000 orders a day, that costs about ₫473M a month (modelled).

The answer

The courier taps one button: No answer, Rescheduled or Refused. Alodi messages the buyer at once, calls if there is silence (60 minutes, then 2 hours, only 08:00–20:30), turns the answer into structured context with a confidence score per field, and hands the next courier a card with when, where and who. Refusals, labels and returns always go to a person.

< 10 s

time-to-context target

measured on stage

≥ 0.90

extraction F1 target

held-out set

416

CS calls a day handled by the agent

projected

₫194M

net saving a month

41% of the loss · projected

02

The problem

Nothing is broken in the route. The context dies in a phone call.

₫40,000

return fee the seller pays per failed order

Shopee §7.1; cap since 1 Apr 2025, GHN · Sourced

8–12%

of orders refused at the door, nationwide

VECOM, Nguoi Lao Dong 19 Jul 2025 · Sourced

0

delivery windows a Shopee buyer can choose

Shopee Help Center 79559 · Sourced

3 in 90 days

failed COD orders before COD is switched off

Shopee Help Center 79196 · Sourced



Field visit · SPX Express hub, Ho Chi Minh City

Why it persists

- **Time is negotiated off the system.** Shopee does not support choosing a delivery time, so buyer and courier agree by phone.
- **One error code replaces the context.** “No answer” or “rescheduled” loses when, where and who; the second attempt repeats the first.
- **People don’t scale.** Phoning 320 failed buyers twice a day for four minutes takes about five full-time callers per area (estimate).
- **The frame is fixed.** Shopee allows 2–3 contact attempts and a reschedule within 5 days (Help Center 79569); anything that helps must fit inside it.

One area, one month

LOSS ITEM	FORMULA	PER MONTH	TYPE
Return fees paid by sellers	4,000 × 8% = 320 failed a day; 75% not rescued × ₫40,000 × 30	₫288.0M	Derived
Courier pay for redelivery	320 × 1.5 attempts × ₫6,000–8,000 × 30	₫86.4M–₫115.2M	Derived
Courier time at unanswered doors	320 × 5 min × courier hourly cost × 30	₫27.4M–₫51.3M	Modelled
CS calling buyers back	320 × 2 calls × 4 min × ₫35,000/h × 30	₫44.8M	Estimate
Total	midpoint of low and high	**₫447M–₫499M · ≈ ₫473M**	Modelled

> Not counted: the seller's lost sale, packaging, ad spend, damaged goods. At VECOM's 12% upper rate every line scales by 1.5×. Full derivation: alodi-numbers.pdf.

03

What we heard: the survey

We asked both sides of the same missed call.

Data status · MOCK 112. The response files in the repository are still the generated placeholders (prep/06-survey, seed 30102026). The values below show what the survey measures and must be replaced by the interview export before they are quoted as findings.

Buyers · n = 100

QUESTION	VALUE
Want to choose a delivery window	86.0%
Called the courier back and couldn't reach them	66.0%
Courier called while busy, often or very often	56.0%
Comfortable with an automated call	58.0%
Would use an Alodi-like service: yes	65.0%

Couriers · n = 12

QUESTION	VALUE
Calls to buyers a day (mean)	107.8
Share of calls unanswered (mean %)	26.6
Buyers call while riding, often	83.3%
Would like an Alodi-like system: yes	83.3%
Minutes saved a day expected (mean)	46.0

Courier sample: SPX Express 4, GHN 2, J&T Express 2, BEST Express 2, GHTK 1, Viettel Post 1.

Instrument: 16 buyer and 15 courier questions in Vietnamese and English (prep/06-survey/codebook.csv). Personal data is dropped before saving (Vietnam PDPL).

04

Alodi

The AI agent that rescues failed deliveries. Autonomous where the rules are clear; human where they aren’t.

The flow, in eight steps

01
Load orders and windows
every morning, per area

02
Ask for a window
≥ 2 options from the route planner

03
Note to the courier
no reply: deliver as planned

04
Deliver, tap the outcome
A / B / Refused, 2-minute undo

05
Flow A · no answer
chat → 60′ call → +2 h call → CS

06
Flow B · rescheduled
10 s memo → ASR → mask → JSON

07
Save the appointment
context card for the next attempt

08
CS decides
a person labels and returns

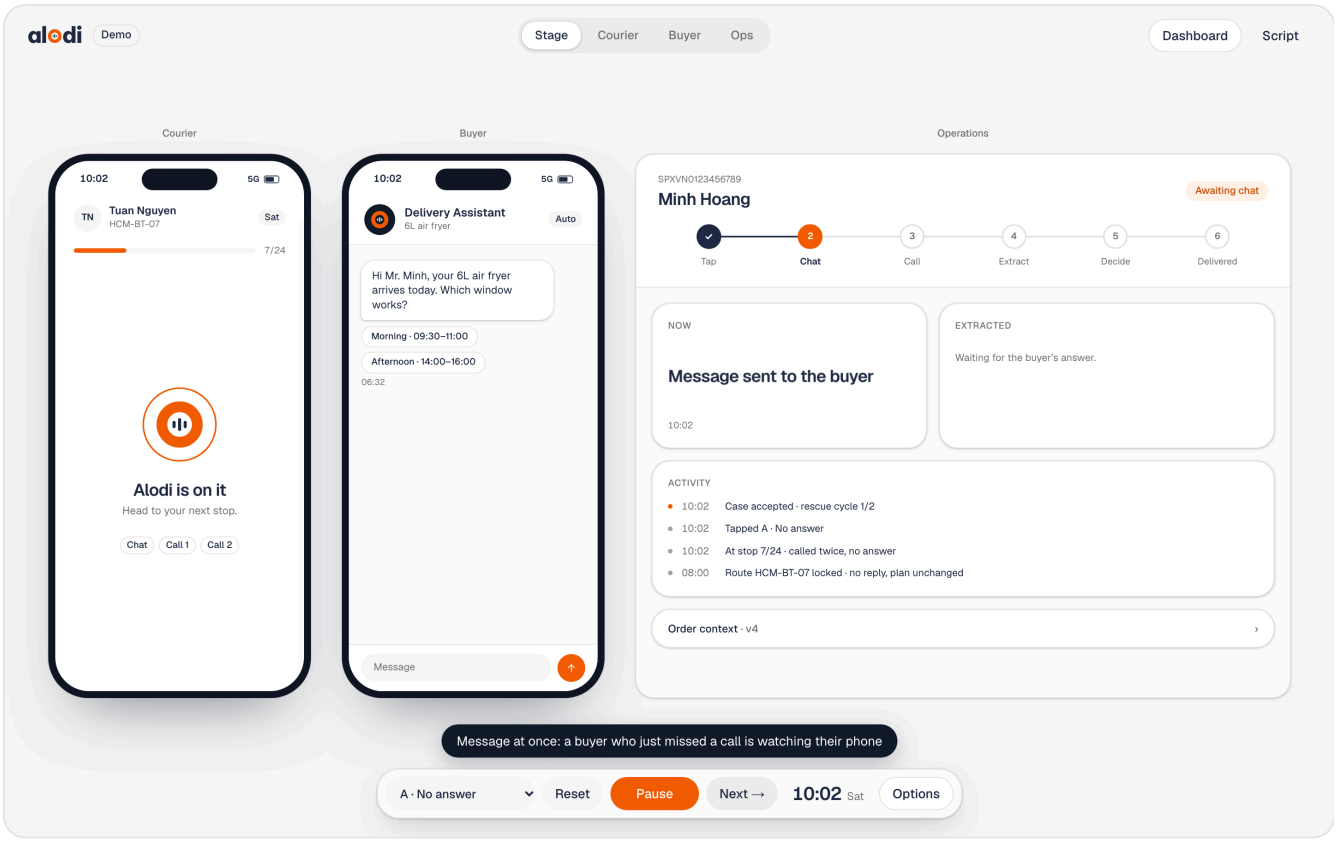
Hard rules (policy guard, in code)

RULE	VALUE
Call hours	08:00–20:30 only; later calls move to the next morning
Calls per cycle	at most 2
Redeliveries	at most 2, inside Shopee’s 5-day frame
Auto-apply	only when every field has confidence ≥ 0.85; otherwise confirm in chat or send to CS
Refused at the door	no buyer contact; straight to CS
Labels and returns	suggested by the agent, approved by a person
Opt-out	“stop” ends all contact
Recordings	consent in the opening line; raw audio kept 30 days

05

The product

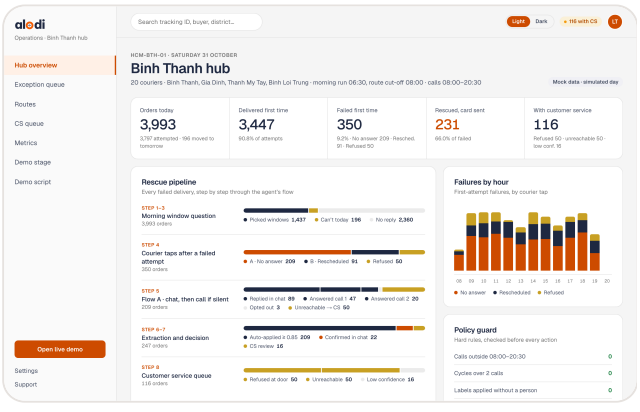
Four screens, one agent, live on stage.



Demo stage: courier app, buyer chat and the operations panel, kept in sync on a fast-forward clock.

Built for the demo

- **Courier app** · one tap, voice memo, context card
- **Buyer channel** · chat with quick replies and the AI call
- **Ops console** · live case, extracted fields, audit trail, CS approval
- **Hub dashboard** · a simulated 4,000-order day: pipeline, failures by hour, policy guard, 20 routes, CS queue
- **Landing page** and **numbers proof** (alodi-numbers.pdf)



Hub dashboard on the simulated day.

3,993
orders in the simulated day
fixed seed · Simulated

350
failed first time · A 209 · B 91 · refused 50
Simulated

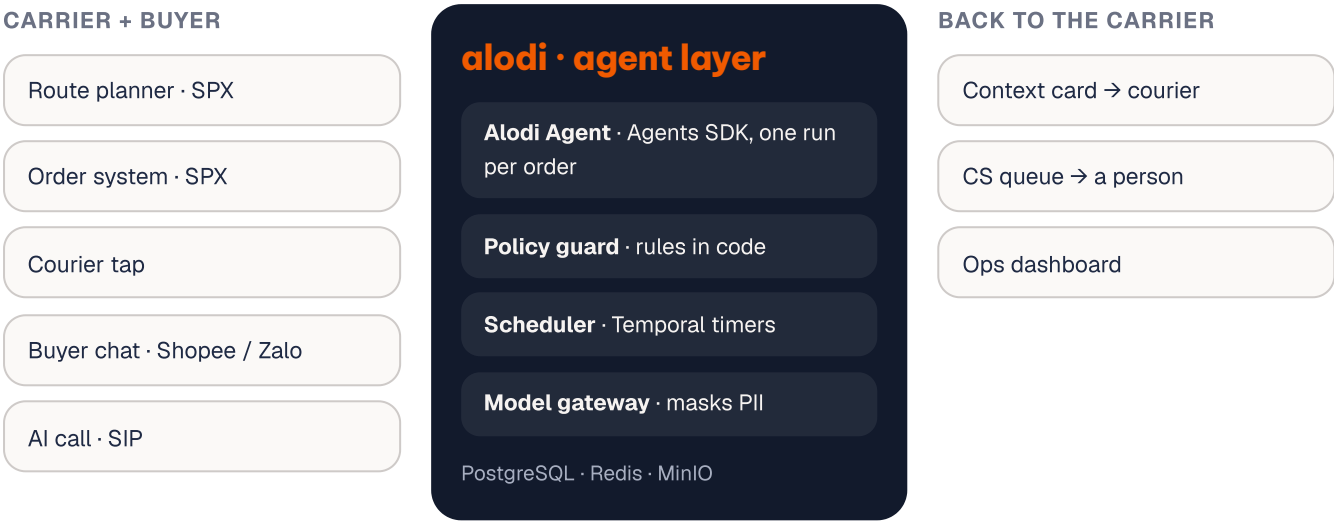
231
rescued with a context card
Simulated

116
handed to customer service
Simulated

06

Architecture

An agent layer between the buyer and the carrier. It replaces nothing and fails open.



COMPONENT	DEMO (31 OCT)	PRODUCTION (PER AREA)
Clients	Next.js PWA on a phone, iPad console, laptop buyer view	SPX driver app, a screen in the SPX dashboard, Shopee Chat / Zalo
API + realtime	FastAPI, REST + WebSocket	FastAPI behind Caddy
Agent	State machine + tool calling	OpenAI Agents SDK, workers on a Temporal queue
Models	OpenAI API, OpenAI TTS, AssemblyAI	gpt-oss-20b on vLLM, PhoWhisper, Piper; OpenAI API as masked fallback
Data	SQLite, recordings folder	PostgreSQL, Redis, Temporal, MinIO (encrypted, 30-day TTL)
Hosting	One VM, Docker Compose	One workstation with two 16 GB GPUs per area; Kubernetes per region at scale

Source: PRD §16, §18.8. The same interfaces run in demo and production; only implementations swap.

07

OpenAI at the core

Runtime on OpenAI models; built and reviewed with Codex.

COMPONENT	ROLE IN ALODI	DEMO	PRODUCTION
gpt-oss-20b	Agent decisions and JSON extraction; runs in about 16 GB with MXFP4	–	Primary, on the area server
Agents SDK	One agent run per order: tools, CS handoff, guardrails, tracing	State machine	Orchestration layer
Responses API	Tool calling and Structured Outputs (strict schema)	Primary	Fallback for ~10% of requests, masked text
gpt-4o-mini-tts	Dynamic phrases inside the AI call	Yes	Short lines only
gpt-4o-transcribe	Second opinion when local ASR is unsure	–	Fallback, masked audio turns
Evals	Graders on the held-out set; gate every change	Codex loop	Release gate
Codex	Specify, build, parallelise, review, evaluate	Yes	Yes

Distributed agent

One orchestrator hands work to six workers (window question, rescue chat, voice, extractor, fallback, guardrails). Each worker is a task on the Temporal queue, so a restart never loses a case, and every worker call passes the policy guard first. Traces go to the events log next to the audit trail.

> Source: PRD §16.7. Production design; the demo runs the OpenAI API behind the same interfaces.

08

Data, privacy and rules

Personal data stays in the area; only extracted JSON leaves it.

- Phone numbers, names and addresses are masked before any model call; only masked text reaches the OpenAI fallback.
- Raw audio and transcripts are encrypted with a 30-day TTL; every change to an order’s context is versioned with the actor and time.
- Recording consent sits in the pre-recorded opening of every call; refusal ends the call and continues by chat or CS.
- Opt-out is checked before every scheduled job; no real number is dialled in the demo (TEST-0001 … TEST-2000).
- Constraints: Vietnam PDPL (from 1 Jan 2026), Decree 53/2022 on data location, Shopee’s 2–3 contacts within 5 days.

09

Evaluation and outcomes

Three numbers measured on stage, one projected.

< 10 s

time-to-context, median

event log · Measured

≥ 0.90

extraction F1 per field

held-out set · Measured

< 2%

wrong auto-applies

eval replay · Measured

0

labels, returns or calls out of hours without a person

Measured

EVAL ITEM	SPEC
Starting set	47 dev samples (35 templated, 12 hard cases)
Final set	≥ 200 samples: chat 45%, call 30%, memo 25%; 30% held out by a different teammate
Scoring	Precision, recall, F1 per field; calibration: ≥ 95% correct above 0.85 confidence
Loop	Codex proposes one change per round; keep it if F1 rises without hurting calibration; stop when held-out stalls
Safe-autonomy run	Replay every sample plus the simulated day; simulation: 209 auto-applied of 350 failures

> Source: PRD §2.3, §23. Numbers shown on today's /metrics page are mock until the eval runs.

10

Economics

One area loses ₫473M a month. Alodi costs ₫21M a month to run.

SCENARIO	CONSERVATIVE	BASE	BEST-IN-CLASS
Rescue rate after a failed attempt (today 25%)	40%	45%	65%
Gross saving a month	₫75.5M	₫119.7M	₫214.7M
Running cost a month	₫20.9M	₫20.9M	₫20.9M
Net saving a month	**₫54.6M**	**₫98.8M**	**₫193.8M**
Net as share of the loss	12%	21%	41%
Net a year	₫0.66B	₫1.19B	₫2.33B

416

CS calls a day handled by the agent

≈ 28 staff-hours · Projected

96

fewer wasted courier trips a day

320 × 0.3 attempts · Projected

24 min

back per courier a day

Projected

₫75M–₫98M

one-time server per area

pays back in ~2 weeks · Listed price + estimate

> Best-in-class uses the top of Shipsy's published 50–65% range (vendor). All savings are projected until a pilot measures Alodi's own rescue rate. Every input and formula: alodi-numbers.pdf.

Competitive landscape

Others reach the buyer. Alodi starts with a meaningful tap and hands back typed context.

	ALODI	CLICKPOST	SHIPSY	AFTERSHIP AGENT	RINGG AI	SHIPROCKET	GHTK
Main market	Vietnam	India	India	Global	India	India	Vietnam
Trigger	Courier taps one meaningful button	NDR code from carrier API	NDR event	Tracking exception	NDR from carrier	Courier marks NDR	Recipient books a slot
Reaching the buyer	Marketplace chat → AI call	SMS, email, WhatsApp, IVR, voice agent	Automated CX Agent	Drafts a message for approval	Voice agent	SMS + IVR (Advanced/Pro)	App, hotline
Asks for a window before delivery	Yes, from the route planner	Not disclosed	Yes	Not disclosed	Not disclosed	Not disclosed	Yes, by half-day
Context for the next attempt	JSON with confidence, versions, audit	Reattempt / return instruction	Not disclosed	Suggested fix	Disposition + instructions	Reattempt / return instruction	Time-window note
Human approval for high-stakes calls	Required; the agent only suggests	Not disclosed	Not disclosed	Yes	Not disclosed	Seller handles it	Call-centre agent
Published results	Measured at the 31 Oct demo	RTO −15–20% (Parth)	Industry rescue 20–30%	−58% resolution time (one customer)	No independent figures	Not disclosed	Not disclosed

> Competitor figures are vendor claims from their own sites (see sources); not independently verified.

12

Roadmap

From a stage demo to a national rollout.

PHASE	WHEN	CONTENT	EXIT CRITERIA
MVP demo	31 Oct 2026	Four screens on mock data	AC-01 to AC-15 pass in rehearsal
Full build	Nov 2026	Server engine, real call audio, ≥ 200 eval samples	Measured metrics on /metrics
Shadow mode	Q1 2027	One area, real events, no buyer contact	Agent vs CS decisions compared; thresholds tuned; DPIA done
Controlled channels	Q2 2027	Real chat and calls for risky orders	0 rule violations; rescue rate measured
Region	H2 2027	5–20 hubs; Vietnamese ASR and TTS self-hosted	Cost per case and latency on target
National	2028	20–100 hubs, GPU pool, second region	Break-even; 99.9% availability

Source: PRD §28.

Risks and assumptions

What could make the numbers wrong, and what we do about it.

RISK	MITIGATION
Buyers ignore unknown callers	Chat first; verified brand name; call only after silence
ASR mishears Vietnamese addresses	Read-back confirmation; 0.85 threshold; PhoWhisper fine-tuning in the pilot
Wrong auto-applied appointment	Threshold plus calibration check; courier can still call; audit log
Wrong label harms a buyer	Labels only by a person
Message cost or spam perception	Opt-out; risky orders only; Shopee Chat before ZNS
Cross-border personal data	PII stays in Vietnam; masked text only to model APIs; self-hosted models at scale
Mock numbers shown as results	Labels mock, simulated, projected on every screen and page

KEY ASSUMPTION	STATUS
One area = 20 couriers, 4,000 orders a day, 200 stops each	Estimate; to check with courier interviews
8% first-attempt failure, 25% baseline rescue	VECOM low end (Sourced); Shipsy (Vendor)
Alodi lifts rescue to 65%	Projected; measured only in a pilot
₫26,000 = $1	Estimate

Source: PRD §26, §27.

Team

Team Develarper.



Nomura Yoshio

Team Lead · Presenter

Product, demo script, pitch



Long Quan Ton

AI Engineer · Solution Architect

Orchestrator, voice agent, eval loop



Khanh Tuong Huynh

AI Engineer · Deployment

Context extractor, ASR, policy guard



Thanh Loi Tran

Data Engineer · Backend

Order Context Object, APIs, metrics log

Sources and verification

Every external figure, where it comes from, and how far we checked it.

SOURCE	WHAT WE USE	WHERE	CHECK
Shopee Return & Refund Policy §7.1–7.2	Seller pays return shipping on failed-delivery orders; none when the carrier is at fault	help.shopee.vn/portal/4/article/77251	Verified 31 Oct 2026 (policy republished 30 Oct 2026, effective 6 Nov 2026; no amount stated)
GHN, who pays for Shopee returns	₫40,000 cap per order since 1 Apr 2025 (₫20,000 express)	ghn.vn/blogs/tip-ban-hang/tra-hang-shopee-ai-chiu-phi	Verified; secondary source quoting Shopee
Nguoi Lao Dong, 19 Jul 2025	VECOM 8–12%, up to 20% remote; Tran Anh Tung 10–15%, 25–30% in sale season	tuoitre.vn/nld/nhuc-nhoi-nan-bom-hang-19625071822023801.htm	Verified
Shopee Help Center 79559	No specific delivery time can be chosen	help.shopee.vn/portal/4/article/79559	Verified
Shopee Help Center 79569	2–3 contacts; reschedule within 5 days	help.shopee.vn/portal/4/article/79569	Verified
Shopee Help Center 79196	3 failed COD orders in 90 days disable COD	help.shopee.vn/portal/4/article/79196	Verified
GHN, courier pay per order	₫6,000–8,000 per delivered order (average)	ghn.vn/blogs/tip-ban-hang/mot-don-hang-shipper-shopee-duoc-bao-nhieu-tien	Verified; not an official Shopee figure
Shipsy, NDR management	Industry rescue 20–30%, best-in-class 50–65%	shipsy.ai/insights/what-is-ndr-non-delivery-report-management	Verified; vendor claim
Metric, E-commerce report 2025	₫429.7T across Shopee, TikTok Shop, Lazada, Tiki; +34.75%	metric.vn (via daibieunhandan.vn, vneconomy.vn)	Verified via press; single data provider
OpenAI, gpt-oss model card	gpt-oss-20b runs within 16 GB with MXFP4	openai.com · Azure AI catalog · vLLM recipes	Verified
Glints Vietnam; Phong Vu; Viettel reseller; oazns.vn; EVN	Courier income, GPU prices, SIP, ZNS and power tariffs	see PRD §18.8	Listed prices, checked 30 Oct 2026 by the team
ClickPost, AfterShip, Ringg AI, Shiprocket, GHTK	Competitor capabilities and results	see web/src/lib/content.ts sources	Vendor claims; not independently verified

> Correction made on 31 Oct 2026: earlier drafts cited a Shopee policy date of 11 Mar 2026 and a VTV article for the ₫40,000 figure. The VTV article describes a promotional support cap, and the policy page has since been republished, so the figure is now cited to the policy (§7.1) plus GHN for the cap.